[INCH-POUND]
A-A-59350
3 August 1999
SUPERSEDING
MIL-T-18134C
29 May 1986

COMMERCIAL ITEM DESCRIPTION

TORCH, WELDING, HAND, OXYACETYLENE, LIGHT DUTY

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-T-18134C for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for a light-duty, hand-held, oxyacetylene-fueled, welding torch to be used for portable welding applications.

2. SALIENT CHARACTERISTICS. In addition to meeting all requirements specified in this document, the torch shall meet the performance requirements of Compressed Gas Association (CGA) standard E-1, "Standard Connections for Regulator Outlets, Torches, and Fitted Hose for Welding and Cutting Equipment", and the performance and labeling requirements of Underwriters Laboratory (UL) standard 123, "Standard for Safety Oxy-Fuel Gas Torches".

2.1 Type. The torch shall be designed for positive-pressure oxyacetylene operation, capable of mixing and controlling the flow of oxygen and acetylene gases to the welding tip.

2.2 Size. The torch shall be sized for hand-held use. Torch length shall be no less than 5 inches and no more than 8 inches. Torch weight shall not exceed 12 ounces.

2.3 Handle. The torch shall include a grip-type handle containing oxygen and acetylene control valves with inlet hose connectors, internal gas passages, and an oxyacetylene outlet connector.

2.4 Valves. The torch shall include oxygen and acetylene control valves which will fully close with no more than 20 inch-pounds of torque applied to the valve knobs and will leak no gas from inlet to outlet under differential pressures up to 50 pounds-per-square-inch gauge (psig). The valve stems shall be sealed with packing to prevent any external gas leakage and to prevent inadvertent stem rotation with up to 4 inch-ounces of torque applied to the knobs. The valve

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AMSC N/A FSC 3433

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knobs shall provide a positive grip for gloved operators and shall not separate from the stem when pulled with 70 pounds of force along the stem axis or loosen from the stem when twisted with 50 inch-pounds of radial torque.

2.5 Gas inlets. The gas inlet hose connections shall be installed at the rear of the torch handle and shall meet the requirements of CGA E-1, connections 020 and 021. Each inlet shall be clearly and permanently marked to identify the oxygen and fuel/acetylene connection points.

2.6 Welding tips. Each torch shall include three welding tips, one in each of the sizes listed in table I. The tip sizes shall be legibly and permanently marked according to the manufacturer's standard practice. Each welding tip shall have a gooseneck shape and shall include either an internal or external mixer. For an external mixer, the tip shall be threaded externally to provide a gas-tight, metal-to-metal fit with the mixing chamber. The welding flame produced, whether long or bulbous, shall be symmetrical around the tip axis.

TABLE I. Welding tip sizes.

Drill Size (Number)	Orifice Diameter (Inches)
73 - 75	0.024 - 0.021
55 - 57	0.052 - 0.043
48 - 51	0.082 - 0.067

2.7 Flashback protection. When operating with the tip and mixing chamber attached and with the oxygen and acetylene set at the manufacturer's recommended pressures, the torch shall withstand five backfires at a rate of at least one per second with no flashback.

2.8 Leakage protection. Joints, packing, O-rings, metal-to-metal mating seals, gas passages, connections, and glands shall withstand internal and external gas pressures of at least 50 psig with no leakage. An internally pressurized control valve, immersed in clean water for five seconds, shall produce no visible bubble, with the valve either closed or opened for internal gas flow. Hand-tightened welding tip seats at maximum operating pressures shall allow no gas leakage.

3. REGULATORY REQUIREMENTS

3.1 Recycovered materials. Offerors and contractors are encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2 Environmental protection. All welding torch products and production methods shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the associated contract. These regulations specify material emissions which are hazardous to the environment or to the user's health and shall be applied during the manufacture, service, transportation, storage, and use of the torch supplied.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The torch provided shall meet the salient characteristics requirements of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market acceptability criteria. The item offered must have been previously sold on the commercial market.

5. PACKAGING

5.1 Preservation, packing, and marking. For acquisition purposes, the torch shall be preserved, packed, and marked as specified in the contract or order.

6. NOTES

6.1 Sources of documents

6.1.1 CGA standards. Copies of CGA standards may be obtained from the Compressed Gas Association, Inc., 1235 Jefferson Davis Highway, Arlington, VA 22202.

6.1.2 UL standards. Copies of UL standards may be obtained from Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.

6.1.3 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

6.2 National stock number (NSN). This CID is associated with NSN 3433-00-255-9332. Other NSNs may also apply to the torch specified in this document.

6.3 Sources of supply. The manufacturers and/or suppliers listed below are known to supply products which meet the salient characteristics requirements of this document. Competition is not limited to the listed firms.

L and M Welding Supply, Inc.
568 Western Drive
Mobile, AL 36607
(334) 470-9997

Uniweld Products, Inc.
2850 Ravenswood Road
Fort Lauderdale, FL 33312-4994
(954) 584-2000

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Navy - SH

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Air Force - 84
Navy - CG, MC, OS

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GSA - 6FET

Preparing activity:
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